

$$\begin{aligned} x + ky &= 1 \\ kx + y &= -1 \end{aligned} \Rightarrow \text{Coeff} \Rightarrow \begin{bmatrix} 1 & k \\ k & 1 \end{bmatrix} \quad (R_2 \rightarrow R_2 - kR_1)$$

echelon coeff $\rightarrow \begin{bmatrix} 1 & k \\ 0 & 1-k^2 \end{bmatrix} \rightarrow \text{Rank (Coeff)} = 1$ if $k^2 = 1$

Aug $\rightarrow \begin{bmatrix} 1 & k & 1 \\ k & 1 & -1 \end{bmatrix} \xrightarrow{R_2 \rightarrow R_2 - kR_1} \begin{bmatrix} 1 & k & 1 \\ 0 & 1-k^2 & -1-k \end{bmatrix}$

$\text{Rank (Coeff)} = 2$ if $k^2 \neq 1$ i.e. $k \neq 1, -1$

$\Rightarrow \begin{bmatrix} 1 & k & 1 \\ 0 & 1-k^2 & -1-k \end{bmatrix} \rightarrow \text{Rank (Aug)} = 2$ if $k^2 \neq 1$

~~$\text{Rank (Aug)} = 1$ if $k = 1$~~

For unique solⁿ

$\text{Rank (Coeff)} = \text{Rank (Aug)} = 2$

$\Rightarrow k^2 \neq 1 \rightarrow k \neq 1 \text{ and } k \neq -1 \rightarrow \text{gives unique sol}^n$

$\hookrightarrow y(1-k^2) = -(k+1)$

$y = \frac{-(k+1)}{1-k^2} \Rightarrow y = \frac{1}{k-1} \quad (k \neq -1)$

$x + ky = 1 \rightarrow x + \frac{k}{k-1} = 1 \rightarrow \boxed{x = \frac{-1}{k-1}}$

$\Rightarrow \vec{x} = \begin{bmatrix} \frac{-1}{k-1} \\ \frac{1}{k-1} \end{bmatrix}$

for no solⁿ

$$\hookrightarrow \text{Rank}(A) \neq \text{Rank}(A|b)$$

$\rightarrow$ Only possible if $\text{Rank}(A) = 1$
and $\text{Rank}(A|b) = 2$

$$\Rightarrow \boxed{k = -1}$$

$\therefore$ For $k = 1 \rightarrow$ no solⁿ

for inf solⁿ,

$$\text{Rank}(A) = \text{Rank}(A|b) = 1$$

$$\Rightarrow \boxed{k = -1} \rightarrow \text{Inf solⁿ}$$

~~✗~~

$$x - y = 1$$

$$y = x - 1$$

$$\text{solⁿ set} \rightarrow (a, a-1) \text{ ie}$$

$$\vec{x} = \begin{bmatrix} a \\ a-1 \end{bmatrix}$$